

Nathan Ma

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EDUCATION

University of Waterloo

BASc, Mechanical Engineering, Option in Artificial Intelligence

Sep 2024 - Present

GPA: 4.0

EXPERIENCE

Mechanical Systems Lead | Goose Robotics

Feb 2026 - Present

- Designing a **12-DOF quadruped robot** with a **PLA/aluminum hybrid frame**, using aluminum structural members for rigid load paths and passive thermal management of onboard electronics.
- Implemented **Watt 6-bar linkages** at each knee to keep servo mass proximal to the body, reducing per-leg rotational inertia and improving dynamic response during locomotion.
- Architected a hierarchical control system with a **Raspberry Pi** for high-level planning communicating over **UART** to a custom **STM32** servo driver generating **12-channel PWM** via HAL timers for low-level joint control.
- Developed per-leg geometric **inverse kinematics** from scratch using trigonometric methods, implementing open-loop **trot gait** with coordinated swing/stance trajectories across diagonal leg pairs.
- Integrating **RPLidar C1** with **ROS2** via the **rplidar_ros** driver, implementing **SLAM Toolbox** for real-time 2D occupancy mapping and onboard environment perception.

Mechanical Engineering Intern | Toyota Motor Manufacturing Canada

Jan 2026 - Apr 2026

- Led a TBP-driven **continuous improvement** project focused on inefficiencies with vision system enabled processes. Reduced shell-body sealer vision system fault downtime by **86%** through process optimization, system revalidation, and coordination with cross-functional production and maintenance teams.
- Designed a **poka-yoke locating guide** for door frame weld jigs via rapid prototyping in **SolidWorks**, constraining part seating geometry to eliminate misload-induced defects without impacting cycle time.
- Coordinated the commissioning, validation, and process optimization of new manufacturing equipment including vision systems, hoists, tooling, and **PLC**-controlled systems to improve cycle time, quality, cost, and safety.
- Calculated new **robotic weld schedules** and retaught robots to reduce weld spatter, minimize weld separation defects, and extend welding tip life.

Industrial Engineering Intern | General Motors

May 2025 - Aug 2025

- Led a **continuous improvement** project focused on eliminating production inefficiencies through targeted **root cause analysis**, collaborating with **cross-functional teams** to optimize line balance. Successfully reduced job cycle times by up to **11.8%** and eliminated significant line downtime, directly resulting in increased shop output.
- Conducted **time studies** and redesigned standardized work across **30+** stations, applying **lean manufacturing** principles to identify wasted movement and ergonomic risks, reducing cycle time and ergonomic strain.
- Updated **10,000+ sqft** of factory layout using **AutoCAD**, streamlining cross-functional operations by delivering precise, standardized floor plans that enhanced coordination and efficiency across departments.
- Observed how **DFMA** decisions influence cost and ergonomics downstream on the plant floor—connecting design with real-world assembly outcomes.

PROJECTS

Internal Cycloidal Actuator | SolidWorks, 3D-Printing, Soldering

- Designed and built a fully integrated **cycloidal actuator** from scratch—custom hand-wound **36S42P outrunner BLDC** motor and internal cycloidal gearbox—modelled as a compact integrated unit in **SolidWorks**.
- Validated electromagnetic design by successfully spinning the motor under **ESC control**; currently integrating **Moteus** driver for closed-loop torque and position control with target **field-oriented control** over CAN bus.
- Diagnosed gearbox seizure from compounding tolerance stack-up due to PLA creep; concluded cycloidal drives exceed FDM precision capabilities and pivoted to a **planetary gearbox** planned top-down from first principles.

TECHNICAL SKILLS

SolidWorks, Siemens NX, CATIA 3DX, KiCad, ANSYS, AutoCAD, Excel, Python, C++, MATLAB, ROS2, Arduino, STM32, ESP32, Embedded Systems, Machining, Welding, GD&T, 3D Printing, Laser Cutting, Soldering, Fusion 360, Autodesk Inventor, FEA (Stress, Thermal, CFD), Topology Optimization, FMEA, Injection Molding, Rapid Prototyping (FDM, SLS, SLA), Silicone Casting, Leak Testing, Thermal Testing, Load Cell Integration, Mechanical Packaging, BOM Management, Materials Selection, Crimping, Wire Routing, Root Cause Analysis, Design of Experiments (DOE), Technical Documentation